Total energy at point A = Total energy at point C

$\frac{1}{2}k{x_A}^2 = mg(x_A + x_C)$

$x_A + x_C = 1.22\ m$

$x_C = 1.02\ m$

(b)(iii) Point of maximum speed is where the system is in equilibrium, net force = 0.

Mg – kx = 0
x = 0.01635 m

By conservation of energy,
Loss in EPE = Gain in KE + Gain in GPE

$\frac{1}{2}k{x_A}^2 - \frac{1}{2}k(0.01635)^2 = Gain\ in\ KE + mg(x_A - 0.01635)$

Gain in KE = 252.95 J

Maximum speed = 4.50 $ms^{-1}$

(b)(iv) Just below B, since x = 0.01635m. At B there is only weight OR at maximum speed elastic force upwards is equal to the weight downwards.

2.

(a)

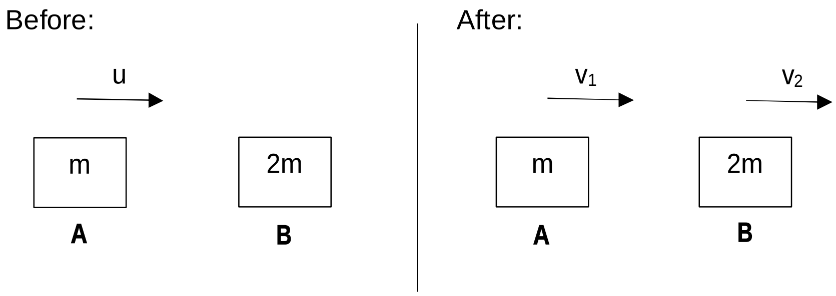
By conservation of energy, the first sphere will accelerate as it loses gravitational potential energy and gains kinetic energy achieving maximum velocity, v, at the lowest position just before it with the second sphere.

Since the spheres are identical, they have the <u>same mass</u>. By conservation of linear momentum, the first sphere with a velocity of v will collide with the next sphere which is at rest. After collision, the second sphere will move off with a velocity v, while the first sphere comes to rest. This continues on till the last sphere moves off with a velocity v.

Since the last sphere has the same velocity as the first sphere, it will rise to the same height as the first sphere where it was released by conservation of energy as the last sphere loses kinetic energy and gains gravitational potential energy.

The last sphere then reaches the same maximum height and comes to rest momentarily. Due to the tension of the string and weight of the sphere, the net force causes the last sphere to accelerate in the opposite direction before colliding with the neighbouring sphere. This motion then repeats itself in an oscillatory manner.

2(b)(i)




By PCLM,

$mu = mv_1 + 2mv_2$ --- (1)

Since collision is elastic, applying relative speed relation,

$u - 0 = v_2 - v_1$ --- (2)

Sub (2) into (1)

$m(v_2 - v_1) = mv_1 + 2mv_2$

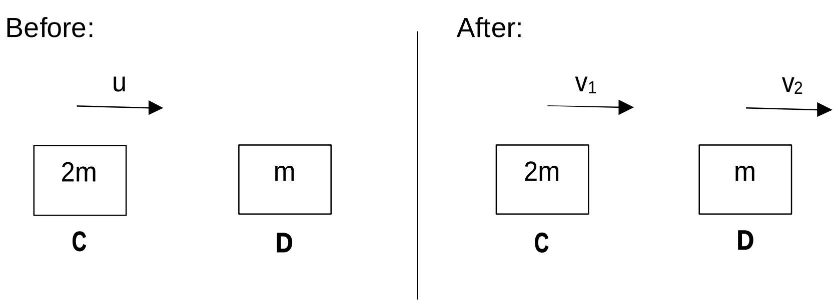
$(v_2 - v_1) = v_1 + 2v_2$

$v_2 = -2v_1$

Hence $v_1 = -\frac{u}{3}$ and $v_2 = \frac{2u}{3}$

$Ratio = \frac{2}{3} = 0.67$

2(b)(ii)




By PCLM,

$2mu = 2mv_1 + mv_2$ --- (1)

Since collision is elastic, applying relative speed relation,

$u - 0 = v_2 - v_1$ --- (2)

Sub (2) into (1)

$2m(v_2 - v_1) = 2mv_1 + mv_2$

$(2v_2 - 2v_1) = 2v_1 + v_2$

$v_2 = 4v_1$

Hence $v_1 = \frac{u}{3}$ and $v_2 = \frac{4u}{3}$

$Ratio = \frac{4}{3} = 1.3$

2(b)(iii)

Collision (i) occurs twice.

Collision (ii) occurs twice.

$$\frac{Max\ vert\ displacement\ of\ first\ sphere}{Max\ vert\ displacement\ of\ last\ sphere} = \frac{Change\ in\ GPE\ of\ first\ sphere}{Change\ in\ GPE\ of\ last\ sphere}$$

By CoE, since the ball comes to rest at max GPE,

$$\frac{Change\ in\ GPE\ of\ first\ sphere}{Change\ in\ GPE\ of\ last\ sphere} = \frac{Max\ KE\ of\ first\ sphere}{Max\ KE\ of\ last\ sphere} = \frac{0.5\, mv_i^2}{0.5\, mv_f^2}$$

Hence

$$\frac{Max\ vert\ displacement\ of\ first\ sphere}{Max\ vert\ displacement\ of\ last\ sphere} = \frac{v_i^2}{v_f^2} = \frac{u^2}{(\left(\frac{2}{3}\right)\left(\frac{4}{3}\right)\left(\frac{2}{3}\right)\left(\frac{4}{3}\right)u)^2}$$

Ratio = 1.60

2(c)

The tension of the spring for all the spheres except for the first sphere is equal to the weight of the sphere since net force = 0. The first sphere will have a greater tension since it undergoes centripetal acceleration, T = mg + $mv^2/r$. Hence extension for the first sphere is greater than that of the rest.

The first sphere and the second sphere no longer collide head-on.

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